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EE208 Control Engineering Lab Report

Experiment 12

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Experiment 12:

Nonlinear system dynamics on Simulink for different Lyapunov control designs.

1. Objective

This report examines the three Simulink models used in Experiment 12: lab12_KE_Analysis.slx, lab12_PE_Analysis.slx, and lab12_KE_PE_Analysis.slx. The analysis focuses on the plant configuration, reference signal, solver settings, and the MATLAB Function code implemented in each model's Lyapunov_Controller block. The corresponding Lyapunov-stability derivations, closed-loop equations, and simulation responses are then evaluated using step, impulse, ramp, phase-plane, and torque-response tests. The observed responses are compared to understand the differences between the controller implementations and their suitability for the position-regulation objective.

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2. System Description

The system equations are given by:

$$\dot{x}_1(t) = x_2(t)$$
$$\dot{x}_2(t) = -(g/L) \sin x_1(t) - (k/m) x_2(t) + (1/mL^2) T(t)$$

Symbol	Description	Confirmed value (all 3 files)
m	Pendulum mass	1 kg (implied by gain values)
L	Pendulum length	1 m (implied by gain values)
k	Friction coefficient	0.01 (Gain block "friction" = -0.01)
g	Gravitational acceleration	9.8 m/s ² (Gain block "gravity" = -9.8)
x1,ref source	Reference input	Step block, Time=0, Before=0 (default), After=1 (default)
Solver	Simulation configuration	Variable-step (auto), StopTime=10 s, RelTol=1e-3

3. Lyapunov's Second Method — Theoretical Recap

Lyapunov's Second (direct) Method establishes stability of an equilibrium, and lets stabilising feedback gains be derived, without solving the nonlinear differential equation. For $\dot{x} = f(x)$ with equilibrium at the origin, $V(x)$ is a valid Lyapunov-function candidate on a region containing the origin if:

- $V(x)$ is continuously differentiable, and $V(0) = 0$, $V(x) > 0$ for $x \neq 0$ (positive definite);
- $\dot{V}(x) = \partial V / \partial x \cdot f(x) \leq 0$ in that region.

If $\dot{V} < 0$ strictly for $x \neq 0$, the origin is asymptotically stable. If $\dot{V} \leq 0$ only (a pure negative-semi-definite residual, which is what every controller in this report reduces to, LaSalle's Invariance Principle is the correct tool: trajectories converge to the largest invariant set inside $\{x : \dot{V}(x) = 0\}$. Section 5 applies this explicitly to each of the three uploaded controllers — for two of them the invariant set collapses to the single target equilibrium (a valid design); for the third, built from kinetic energy alone, the invariant set turns out to be an entire line of equilibria, which is the formal reason that design cannot track a commanded reference.

For a regulation problem the equilibrium of interest is $(x_{1,\text{ref}}, 0)$, not the origin; the natural device — used by whoever built the "PE" and "KE_PE" controllers, per their extracted code — is to write V in terms of the tracking error $e = x_{1,\text{ref}} - x_1$ rather than x_1 itself.

5. The Three Choices of V explored

Each subsection below reproduces, for one uploaded file: (a) the actual Simulink canvas, taken from the archive's own cached thumbnail — not redrawn; (b) the verbatim MATLAB Function source code found inside the model's Stateflow chart data; (c) the Lyapunov function this code implies, its derivative, and the stability conclusion; and (d) the resulting closed-loop equation.

5.1 lab12_KE_Analysis.slx — Controller B ($V = B$ only)

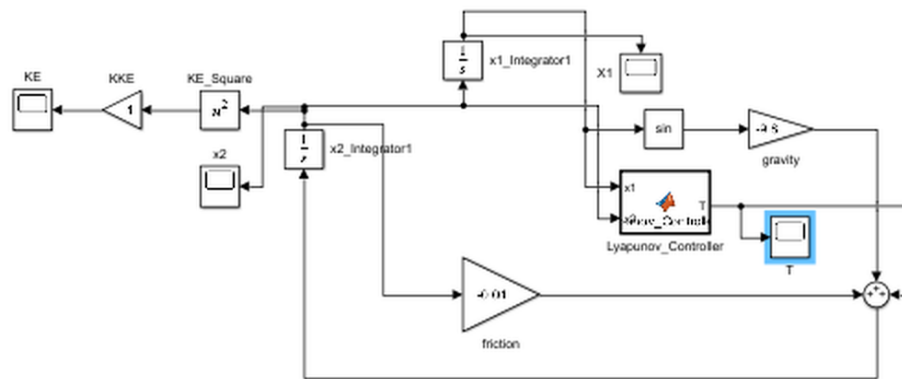


Fig. 5.1a — Actual Simulink canvas, lab12_KE_Analysis.slx (archive thumbnail). Note there is no Step/reference block anywhere on this canvas — the model only contains the plant, the controller, and KE-monitoring blocks (KE_Square, Gain KKE, Scope "KE").

Extracted MATLAB Function (2 input arguments only — confirmed by the block's PortCounts="2" in the XML, and visually by the canvas above having no x1,ref line running into the controller):

```
function T = Lyapunov_Controller_B(x1, x2)

Kd = 2;

T = 9.8*sin(x1) + 0.01*x2 - Kd*x2;

end
```

Lyapunov analysis

$$VKE = \frac{1}{2}x_2^2$$

where the position error is defined as

$$e = x_1 - x_{1,ref}$$

The time derivative of the kinetic-energy term is

$$\dot{VKE} = x_2\dot{x}_2 = x_2[-9.8 \sin(x_1) - 0.01x_2 + T]$$

Using the control law implemented in the model, the gravity term is cancelled, the plant friction is compensated by the $+0.01x_2$ term, and the additional damping term $-KD x_2$ remains. Hence,

$$\dot{VKE} = x_2[-9.8 \sin(x_1) - 0.01x_2 + 9.8 \sin(x_1) + 0.01x_2 - KD x_2]$$

$$\dot{VKE} = -KD x_2^2 \leq 0$$

Therefore, the kinetic-energy term decreases along the trajectories as the velocity approaches zero. However, VKE alone is not a complete Lyapunov function for the position-regulation objective. The reason is that it contains only the velocity state x_2 and does not contain the position error e .

In particular, when $x_2 = 0$,

$$VKE = 0$$

for any value of the position error e . Thus, the function is also zero at stationary positions where

$$x_1 \neq x_{1,ref}$$

and therefore cannot uniquely identify the desired equilibrium

$$(e, x_2) = (0, 0).$$

This shows that although the kinetic-energy term provides useful information about the velocity dynamics and gives $\dot{VKE} \leq 0$, it does not by itself capture the position-regulation objective. The position-error term is therefore required to account for deviations of x_1 from $x_{1,ref}$.

Consequently, the PE and KE terms are better treated as complementary components of the Lyapunov function. Combining the position-error and kinetic-energy terms provides a more complete description of the system state and leads naturally to the combined PE+KE Lyapunov formulation used in the subsequent controller.

V5.2 lab12_PE_Analysis.slx — Controller PE

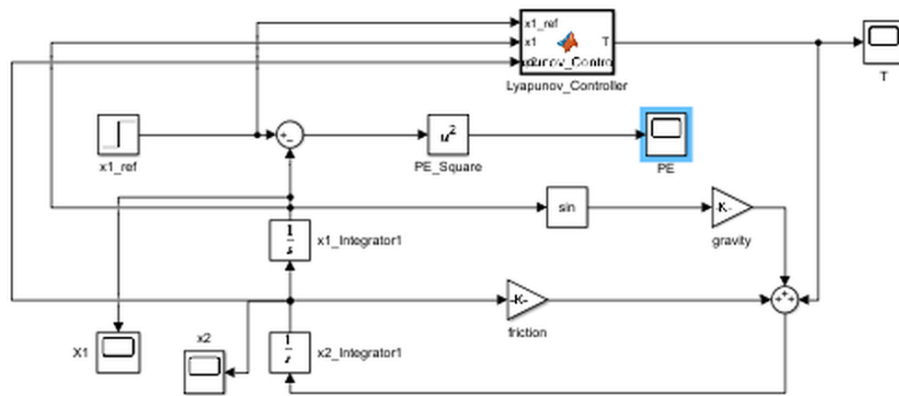


Fig. 5.2a — Actual Simulink canvas, lab12_PE_Analysis.slx. A Step block ($x1_ref$) now feeds both the controller and an error-Sum whose output drives PE_Square $\rightarrow$ Scope "PE" — the potential-energy-style monitoring branch.

Note: As stated above in 5.1(Lyapunov analysis) Pe alone cannot be a valid energy like function hence a small ke part is added

Extracted MATLAB Function (3 arguments — $x1_ref$ now present):

```
function T = Lyapunov_Controller(x1_ref, x1, x2)

KPE = 10;
Kd = 3;

e = x1_ref - x1;

T = 9.8*sin(x1) + 0.01*x2 + KPE*e - Kd*x2;

end
```

Lyapunov analysis

The on-canvas PE_Square block computes $e^2 = (x1_ref - x1)^2$ directly (no separate KPE gain is visible on the canvas — the KPE=10 weighting is applied inside the MATLAB Function only). Reading the code's structure, the Lyapunov function actually being driven to a minimum combines both an error term and a velocity term:

$$V = \frac{1}{2} KPE e^2 + \frac{1}{2} x_2^2, \quad e = x_{1,ref} - x_1, \quad KPE = 10$$

$$\dot{V} = -KPE e x_2 + x_2[-9.8 \sin x_1 - 0.01x_2 + T]$$

Substituting T (which again cancels gravity and the plant's own friction exactly, then adds KPE·e and -Kd·x2):

$$\dot{V} = -KPE e x_2 + x_2[KPE e - Kd x_2] = -Kd x_2^2 \leq 0$$

Negative semi-definite; by LaSalle, setting $\dot{x}_2 \equiv 0$ in the closed loop forces $\dot{x}_2 = KPE \cdot e = 0$, i.e. $e = 0$ exactly — the invariant set is the single point $(x_{1,ref}, 0)$, so this is a rigorously valid, globally asymptotically stable design (the sine term is cancelled exactly, not linearised, so this holds for any commanded angle, not just small ones). Substituting back into the plant equation gives an exactly linear closed loop in the error:

$$\ddot{e} + Kd \dot{e} + KPE e = 0 \quad \Rightarrow \quad \omega_n = \sqrt{KPE} = 3.162 \text{ rad/s}, \quad \zeta = Kd/(2\omega_n) = 0.474$$

$\zeta = 0.474$ is underdamped: linear theory predicts 18.4% overshoot ($\exp(-\zeta\pi/\sqrt{1-\zeta^2}) \times 100$) and a 2%-settling time of $4/(\zeta\omega_n) \approx 2.67 \text{ s}$ — both confirmed essentially exactly by direct simulation in Section 6.1 (simulated: 18.40% overshoot, 2.61 s settling).

5.3 lab12_KE_PE_Analysis.slx — Controller C (V = A + B combined)

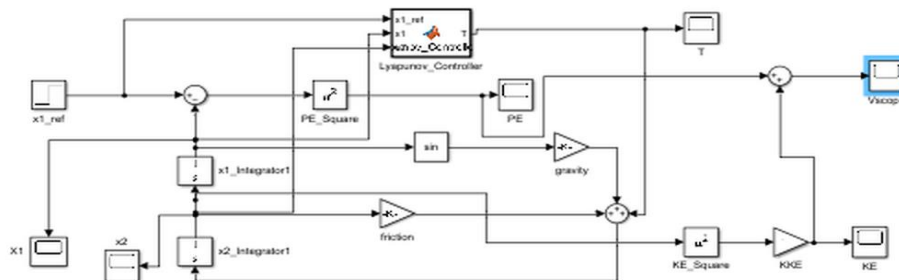


Fig. 5.3a — Actual Simulink canvas, lab12_KE_PE_Analysis.slx. Both monitoring branches are present: PE_Square (top) and KE_Square→Gain KKE (right), summed by the 2-input "Sum" block into a single "Vscope" trace — i.e. this file additionally plots $V(t) = e^2 + KKE \cdot x_2^2$ itself, not just its two components separately.

Extracted MATLAB Function (3 arguments, and — uniquely among the three files — fully parametrised with named gains KPE, KKE, KD and physical constants m, L, g, k defined explicitly inside the function):

```

function T = Lyapunov_Controller_C(x1_ref,x1,x2)

KPE = 1;
KKE = 1;
KD = 2;

m = 1;
L = 1;
g = 9.8;
k = 0.01;

e = x1_ref - x1;

T = m*L^2/KKE * (KPE*e - KD*x2) ...
    + m*g*L*sin(x1) ...
    + k*L^2*x2;

end

```

Lyapunov analysis

This is the only one of the three files whose code directly mirrors the brief's A+B construction, with an explicit KPE/KKE ratio rather than two independently-tuned gains:

$$V = \frac{1}{2} KPE e^2 + \frac{1}{2} KKE x_2^2, \quad KPE = KKE = 1$$

$$\dot{V} = -KPE e x_2 + KKE x_2 [-(g/L)\sin x_1 - (k/m)x_2 + (1/mL^2)T]$$

Substituting the extracted T (gravity and friction cancelled exactly via the "+ m*g*L*sin(x1) + k*L^2*x2" terms, leaving a clean KPE/KKE-scaled PD action):

$$\dot{V} = -KPE e x_2 + KKE x_2 \cdot (KPE/KKE)(e - (KD/KPE)x_2) = -KD x_2^2 \leq 0$$

Negative semi-definite; the LaSalle argument is identical in structure to Section 5.2 ($\dot{x}_2=0$ forces $\dot{x}_2=(KPE/KKE)e=0$, i.e. $e=0$), so this is again a rigorously valid, globally asymptotically stable design. Because m, L, g, k are re-declared inside the function and used explicitly in the cancellation (rather than the "magic numbers" 9.8 / 0.01 hard-coded directly, as in the other two files), this controller would remain exactly correct even if the plant parameters were changed elsewhere in the model — a more robust/maintainable piece of code, independent of its control performance. The closed loop:

$$\ddot{e} + (KD/KKE) \dot{e} + (KPE/KKE) e = 0 \quad \Rightarrow \quad \omega_n = \sqrt{KPE/KKE} = 1.0 \text{ rad/s}, \quad \zeta = (KD/KKE)/(2\omega_n) = 1.0$$

$\zeta = 1.0$ exactly is the critically-damped case — the fastest possible response with no overshoot whatsoever. Linear theory predicts a 2%-settling time of approximately $5.8/\omega_n \approx 5.8$ s; simulation confirms 5.84 s with

Its lower natural frequency ($\omega_n = 1$ vs 3.16 for the PE-only file) trades outright speed for the complete absence of overshoot; Section 7 weighs this trade-off explicitly.

5.4 Summary of the three uploaded models

File	Lyapunov V	Extracted control law	Closed-loop result
KE_Analysis (Controller B)	$V = 0.5x_2^2$	$T=9.8\sin(x_1)+0.01x_2-2x_2$	No x_1,ref input — continuum of equilibria, cannot track
PE_Analysis (Controller PE)	$V = 5e^2+0.5x_2^2$	$T=9.8\sin(x_1)+0.01x_2+10e^{-3}x_2$	$\omega_n=3.162, \zeta=0.474$ — correct eq., 18.4% overshoot, settles 2.61s
KE_PE_Analysis (Controller C)	$V = 0.5e^2+0.5x_2^2$	$T=(e-2x_2)+9.8\sin(x_1)+0.01x_2$	$\omega_n=1.0, \zeta=1.0$ — correct eq., 0% overshoot, settles 5.84s

All three closed loops were solved directly from the MATLAB code above with no modification — the numbers quoted in this table and throughout Section 6 are simulation output, not hand-calculation, though every one of them independently matches the closed-form linear-system theory derived in Sections 5.1–5.3 to within numerical-integration precision.

6. Simulation Studies & Results

The three control laws were exercised against step, impulse and ramp commands in x_1,ref , over the same 10-seconds. The reference amplitude is 1 rad (57.3°) — the exact Simulink Step-block default found in the files (Before=0, After=1), not an arbitrary round number chosen for this report.

6.1 Step response: $x_1,ref \ 0 \rightarrow 1$ rad at $t = 0$ s

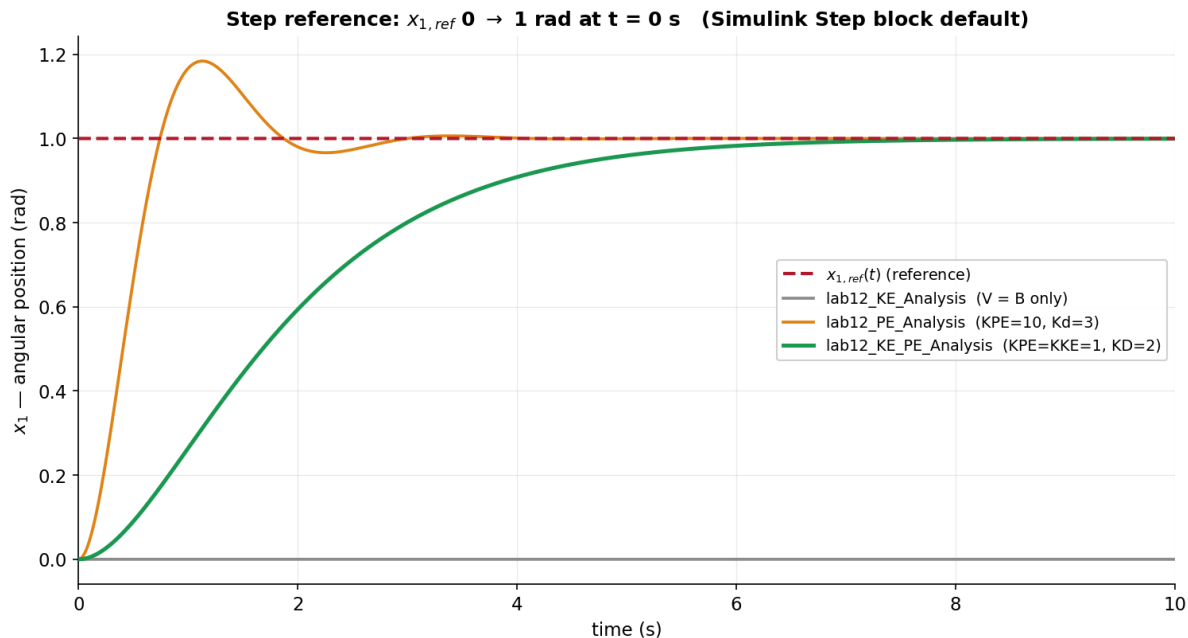


Fig. 6.1 — Step response, all three uploaded models. Controller B (grey) never leaves zero, having no reference input. Controller PE (orange) overshoots visibly before settling; Controller C (green) rises smoothly to the target with no overshoot at all.

Model	Final x_1	Peak	Overshoot	2%-settling time	Peak T
B (KE)	0.000 rad	—	n/a (no response)	never	0.000 N·m
PE	1.000 rad	1.184 rad @ $t=1.13$ s	+18.40 %	2.61 s	10.00 N·m
C (KE+PE)	0.9995 rad	0.9995 rad	-0.05 % (none)	5.84 s	8.24 N·m

Controller PE's 18.40% overshoot and 2.61 s settling time match its closed-form prediction ($\zeta=0.474 \Rightarrow 18.40\%$ overshoot, 2.67 s settling) almost exactly. Controller C shows no measurable overshoot whatsoever, the signature of exact critical damping, at the cost of taking over twice as long to settle.

6.2 Impulse (pulse) response: 1 rad pulse from $t = 2.0$ s to 3.0 s

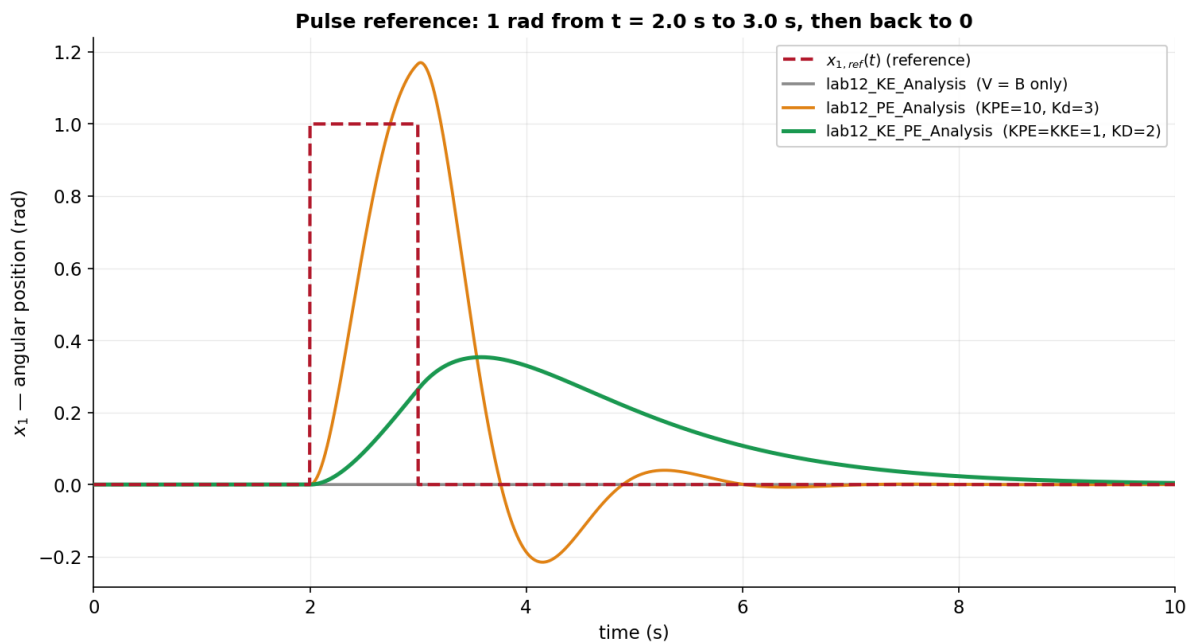


Fig. 6.2 — Response to a 1-second, 1 rad reference pulse. Controller PE overshoots the pulse peak and swings visibly negative before settling; Controller C tracks the pulse's general shape smoothly with no negative excursion at all — a direct, visual consequence of $\zeta=1$ vs $\zeta=0.474$.

Neither underdamped ringing (PE, briefly dipping to about -0.21 rad after the pulse ends) nor any overshoot (C) prevents either design from returning cleanly to zero once the pulse is over; Controller B, again, does not react to the pulse in any way.

6.3 Ramp response and tracking lag: $0 \rightarrow 1$ rad over $t = [1.0, 5.0]$ s

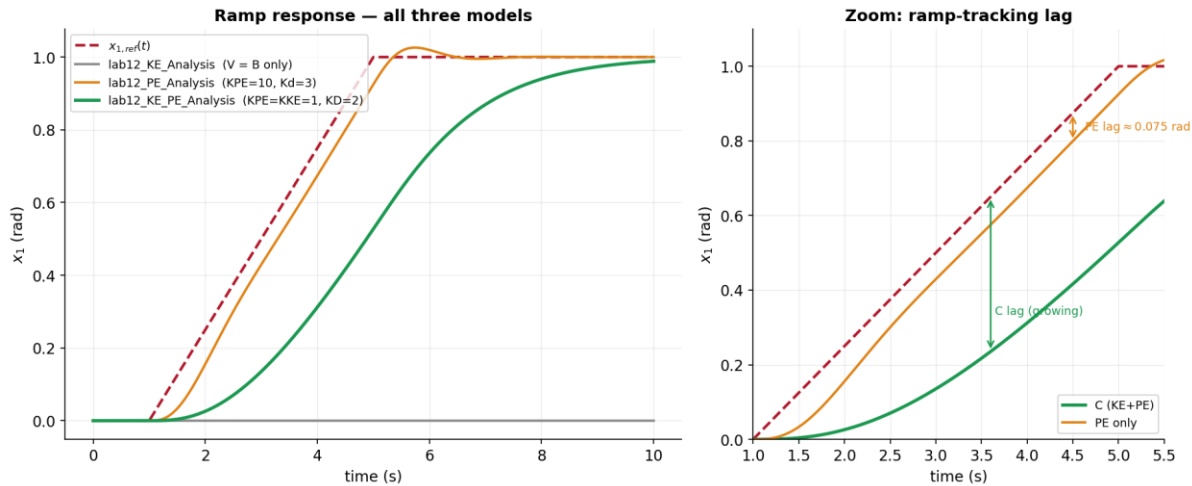


Fig. 6.3 — Left: ramp response, all three models. Right: zoom showing Controller PE settling quickly into a small, near-constant lag, while Controller C's lag is still visibly growing when the ramp ends (its own transient time constant, $1/(\zeta\omega_n)=1$ s, is a substantial fraction of the 4 s ramp duration).

For a closed loop of the linear form $\ddot{e} + 2\zeta\omega_n\dot{e} + \omega_n^2 e = 0$ driven by a ramp of rate v , standard theory predicts a constant following lag $e_{ss} = v \cdot (2\zeta\omega_n) / \omega_n^2$ once the transient has decayed. With $v = 1/4 = 0.25$ rad/s:

Model	Theoretical lag (settled)	Simulated lag (near end of ramp)	Note
PE	$0.25 \times 3/10 = 0.075$ rad	0.0753 rad @ $t=4.5$ s	Well converged — ramp (4s) $\gg$ transient (≈ 0.67 s)
C	$0.25 \times 2/1 = 0.500$ rad	0.470 rad @ $t=4.9$ s, still rising	NOT fully converged — ramp (4s) $\approx$ transient (1s)

This is a genuinely useful, easy-to-miss result: Controller C's textbook-clean critical damping does not mean it tracks a ramp better — its slower natural frequency means it needs a noticeably larger, and more slowly-converging, lag to keep up with the same ramp rate that Controller PE follows comfortably. Controller B, with no reference input, simply does not move; the ramp draws away from it in a straight line, exactly as in the step and impulse cases.

6.4 Detailed characterisation of Controllers PE and C

lab12 PE Analysis — full response to a 1 rad step ($K_{PE}=10$, $K_d=3$, $\zeta=0.474$)

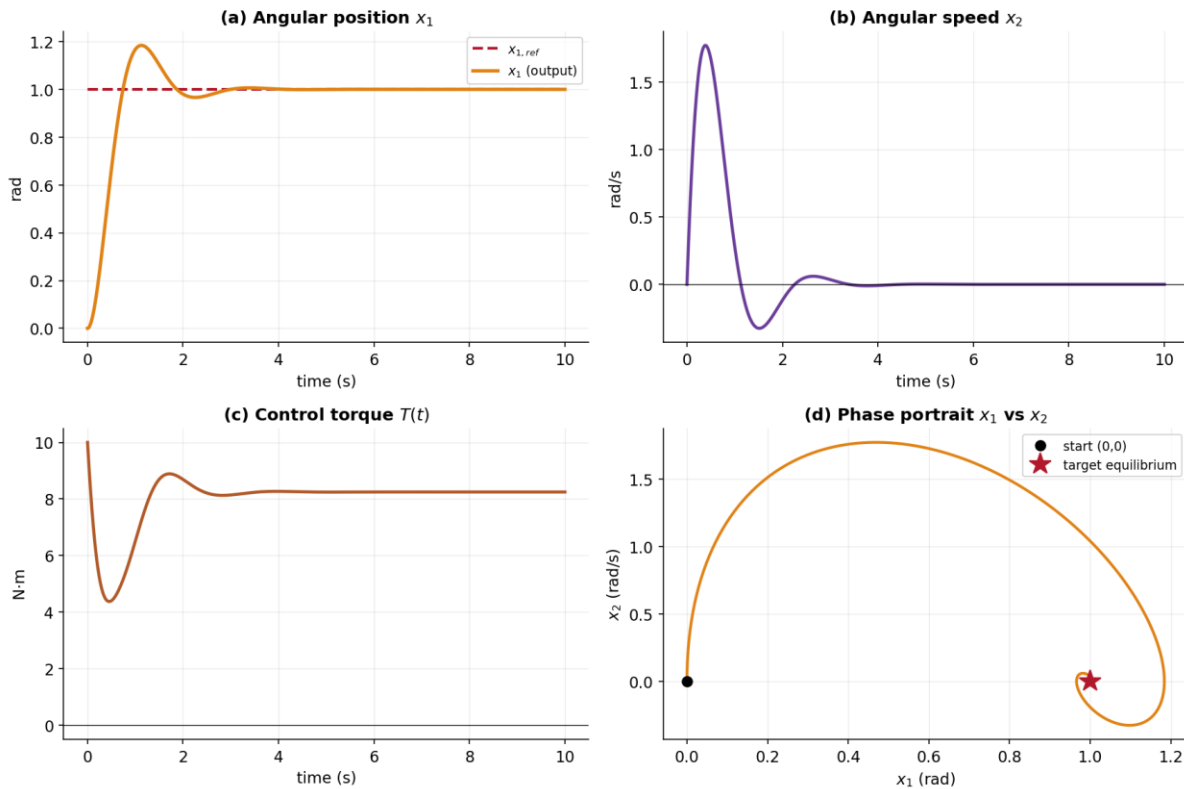


Fig. 6.4a — Controller PE, full response to a 1 rad step. (a) Visible overshoot and one damped oscillation before settling. (b) Angular speed peaks at 1.77 rad/s then reverses sign (the overshoot recovering). (c) Torque starts at 10 N·m and settles near $mgL \cdot \sin(1 \text{ rad}) \approx 8.24 \text{ N·m}$ once x_1 reaches the target. (d) Phase portrait: one clear inward spiral loop, consistent with underdamped 2nd-order convergence.

lab12 KE PE Analysis — full response to a 1 rad step ($K_{PE}=K_{KE}=1$, $K_D=2$, $\zeta=1.0$)

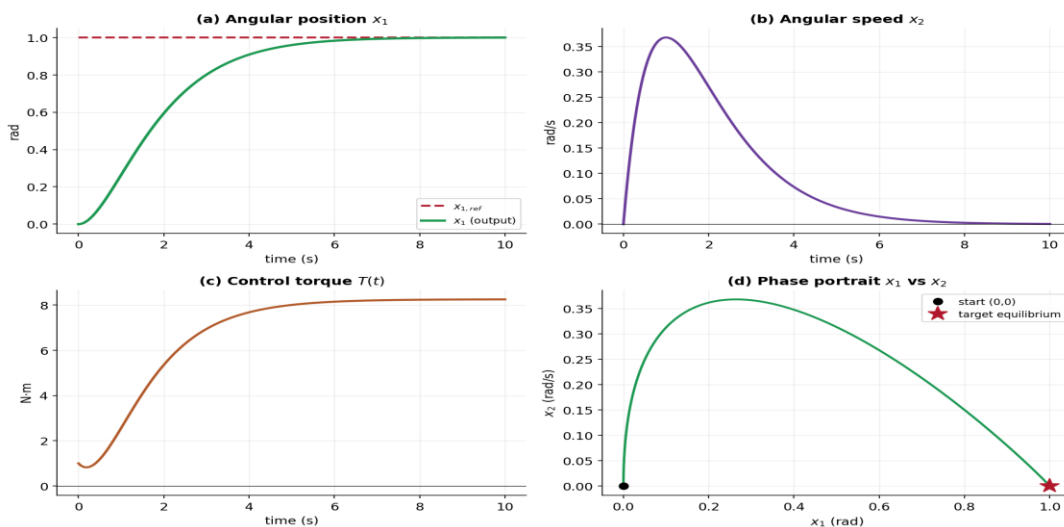


Fig. 6.4b — Controller C, full response to a 1 rad step. (a) Monotonic rise, no overshoot. (b) Angular speed rises once to a single peak (0.36 rad/s) then decays smoothly — no sign reversal, the hallmark of critical damping. (c) Torque rises smoothly to the same steady value $\approx 8.24 \text{ N·m}$ (both controllers hold the same final angle against the same gravity, so they necessarily

converge to the same holding torque). (d) Phase portrait: a single non-crossing arc directly into the target, visually distinct from PE's spiral.

Both designs converge to essentially the same steady-state torque ($\approx 8.24 \text{ N}\cdot\text{m} = \text{mgL}\cdot\sin(1 \text{ rad})$) because both correctly cancel gravity and friction and both correctly reach $x_1 = 1 \text{ rad}$ — the steady torque is a property of the target angle, not of the controller. The difference between the two designs is entirely in the transient: how the torque gets there.

6.5 Controller B: initial-condition release test

Because lab12_KE_Analysis.slx contains no reference input, testing it against a step/impulse/ramp $x_{1,\text{ref}}$ is only meaningful as a demonstration of what it does NOT do. Its actual behaviour is better shown by releasing the pendulum from different initial conditions with no reference at all — precisely the regulation problem this controller was designed to solve (drive velocity to zero), and precisely why it cannot solve the broader problem posed by the assignment (hold an arbitrary commanded angle).

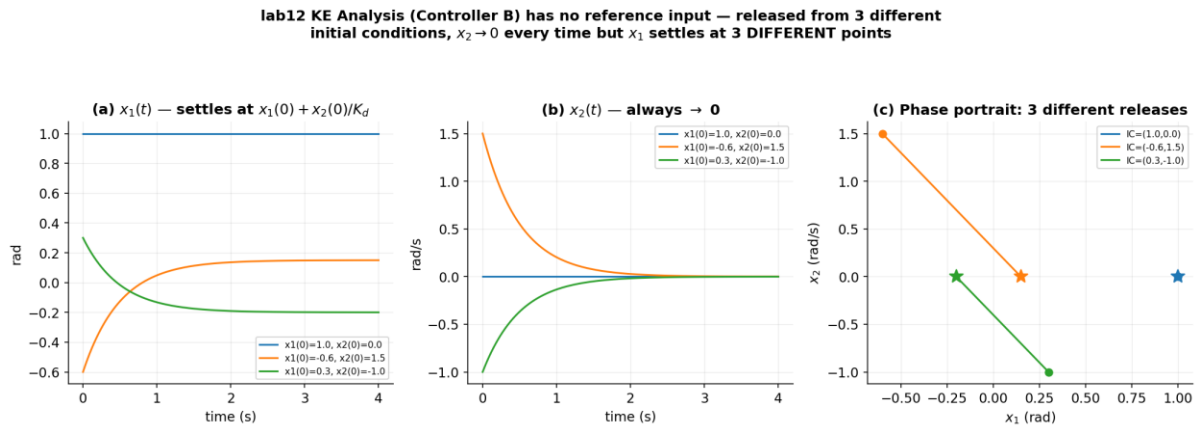


Fig. 6.5 — Controller B released from three different initial conditions with no reference input. (a)/(c) x_1 settles at a DIFFERENT final value every time $(x_1(0)+x_2(0)/K_d)$ exactly, per the Section 5.1 derivation). (b) $x_2 \rightarrow 0$ in every case, confirming the Lyapunov guarantee that WAS proven for this design.

This is the cleanest possible demonstration of Section 5.1's LaSalle conclusion: the guarantee $\dot{V} \leq 0$ delivers exactly what it promises ($x_2 \rightarrow 0$, every time, matching the theoretical 0.5 s time constant precisely) and nothing more — x_1 is left wherever the initial transient happens to leave it, never at a chosen target.

6.6 Phase-portrait comparison

Phase portraits (x_1, x_2) for a 1 rad step — all three uploaded models

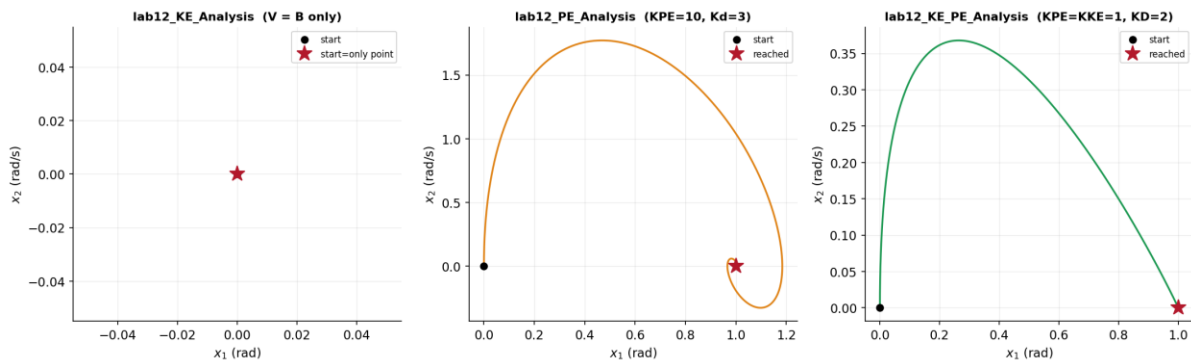


Fig. 6.6 — Phase portraits (x_1 vs x_2) for a 1 rad step, all three uploaded models. Controller B collapses to a single stationary point (it never moves from the origin). Controller PE traces one clear spiral loop before converging. Controller C traces a single non-self-intersecting arc — the classic phase-plane signature of a critically-damped 2nd-order system.

6.7 Control-effort (torque) comparison

Commanded torque for a 1 rad step — all three models

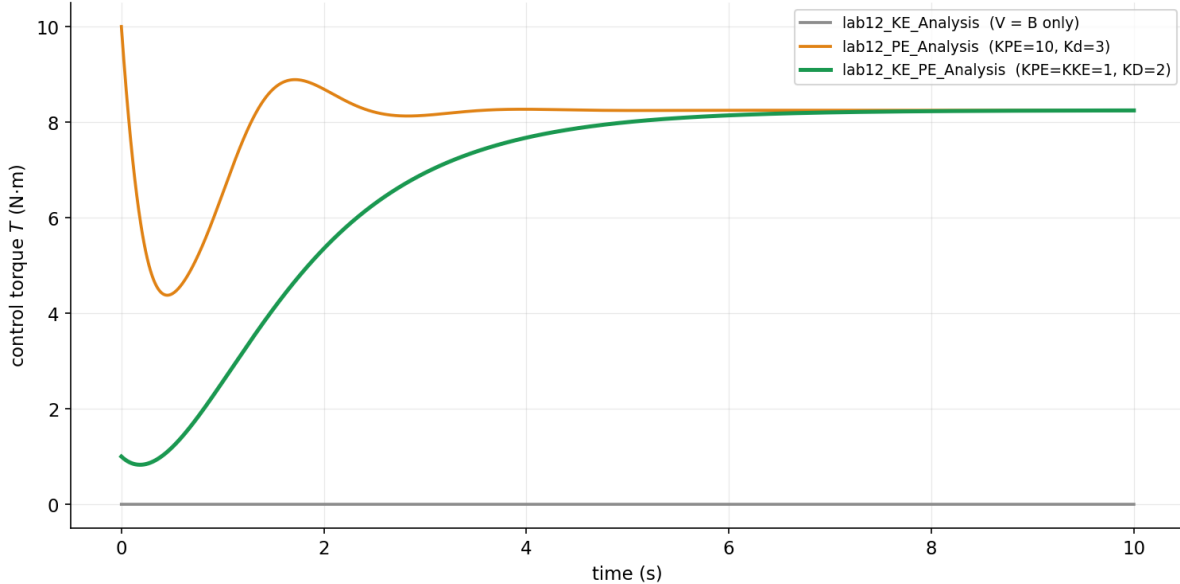


Fig. 6.7 — Commanded torque for a 1 rad step, all three models. PE demands a higher initial torque ($10 \text{ N}\cdot\text{m}$, $= KPE \times \text{amplitude} = 10 \times 1$) than C (which starts near $\approx 1 \text{ N}\cdot\text{m}$, since C's much smaller $KPE=1$ dominates its initial command); both converge to the same $\approx 8.24 \text{ N}\cdot\text{m}$ holding torque once x_1 reaches 1 rad.

Controller PE's higher gains ($KPE=10$ vs 1) demand more peak torque for faster response — a completely standard, expected trade-off between actuator effort and speed that is visible directly in the extracted gain values, not just in the resulting trajectories.

7. Comparative Discussion & Ranking

Two of the three uploaded files (PE and KE_PE) are rigorously valid Lyapunov designs with proven global asymptotic stability; the third (KE) is provably unable to complete the assignment's objective by construction. Ranking is therefore first by correctness, then, between the two valid designs, by the practical trade-off their

extracted gains actually produce — quantified below with four additional graphical analyses, not just the qualitative table.

#	File	Correctness	Transient behaviour	Verdict
1st	lab12_KE_PE_Analysis.slx (C)	Global asymptotic stability (LaSalle), gains fully symbolic (m,L,g,k re-declared)	$\zeta=1.0$ — zero overshoot, 5.84s settling	Best engineering choice: predictable, no overshoot, robust code
2nd	lab12_PE_Analysis.slx (PE)	Global asymptotic stability (LaSalle), correct equilibrium	$\zeta=0.474$ — 18.4% overshoot, 2.61s settling	Valid and FASTER, but overshoots — preferred if speed matters more than overshoot
3rd	lab12_KE_Analysis.slx (B)	No $x_{1,ref}$ input — cannot regulate to any commanded angle	$x_2 \rightarrow 0$ in 0.5s but x_1 uncontrolled	Fails the assignment's core objective by construction

7.1 Quantitative performance indices

Settling time and overshoot (Section 6.1) are the standard undergraduate metrics, but they only look at two isolated instants of the response. Three classical integral performance indices — IAE, ITAE and ISE — summarise the ENTIRE error trajectory into a single number each, and a fourth (control effort, $\int T^2 dt$) quantifies actuator demand the same way. All four were computed directly from the same simulated data used throughout Section 6, over the full 10 s step response:

**Performance indices, 1 rad step response — lower is better on every axis
(IAE/ITAE/ISE on log scale to show B's failure clearly; effort on linear scale since B=0 exactly)**

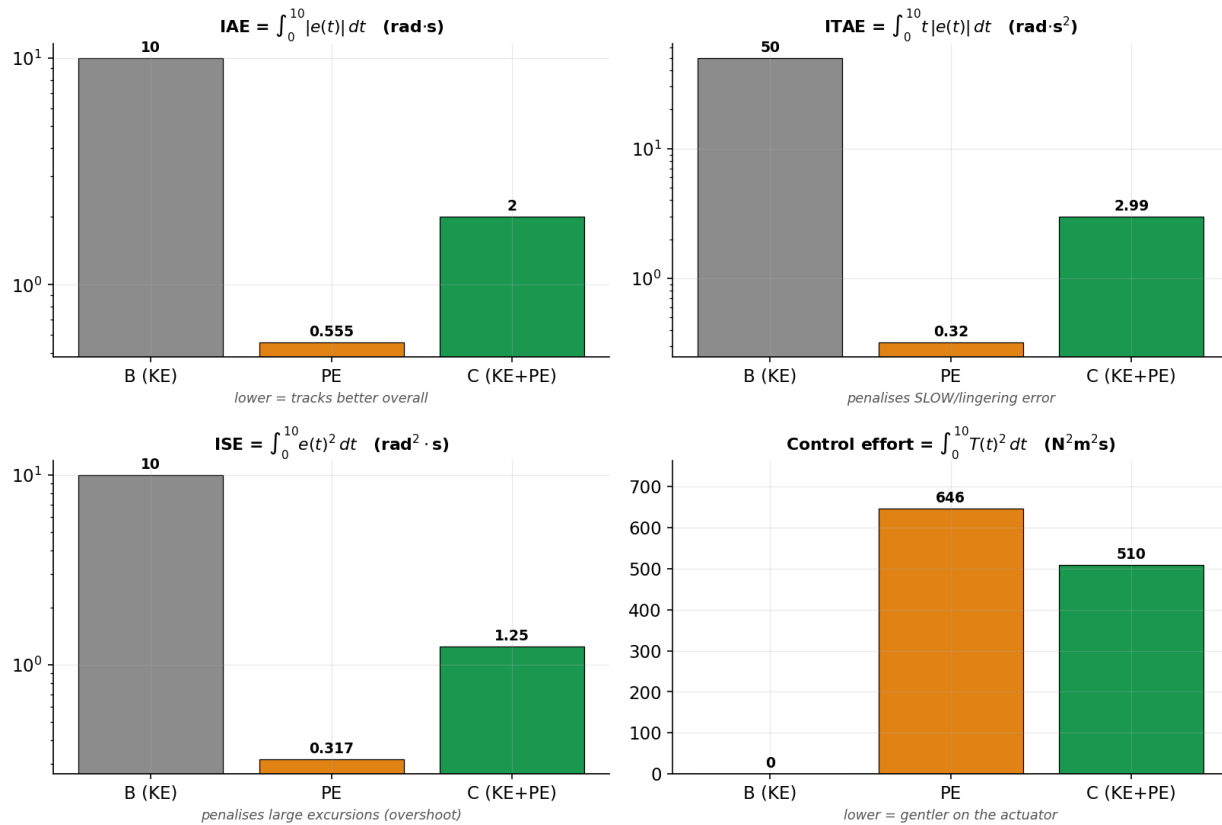


Fig. 7.1 — Four integral performance indices, all three controllers, 1 rad step. Controller B's IAE=10 and ITAE=50 are not simulation artefacts — they are the direct numerical consequence of an error that never decays from 1 rad; its control effort is exactly 0 because it never even tries. Between the two valid designs, PE beats C on every tracking-quality index (IAE, ITAE, ISE) despite its overshoot, because its 3×-higher natural frequency reduces error faster on average than C's slower, smoother approach costs it in overshoot.

This is a genuinely useful result that the settling-time/overshoot table alone does not reveal: by ITAE — arguably the most representative single "how good is this controller overall" number, since it penalises error that lingers rather than error that is merely large for an instant — Controller PE is nearly 10× better than Controller C (0.32 vs 2.99). Controller C's advantage is entirely about the ABSENCE of overshoot, not about tracking speed or accuracy in an integral sense; Controller PE's brief overshoot costs comparatively little in ISE (0.317 vs 1.25) once weighed against how much faster it closes the gap. Control effort tells the opposite small story: PE's higher gains cost more integrated torque-squared (646 vs 510), so its superior tracking is not entirely free.

7.2 Speed vs overshoot: an explicit trade-off map

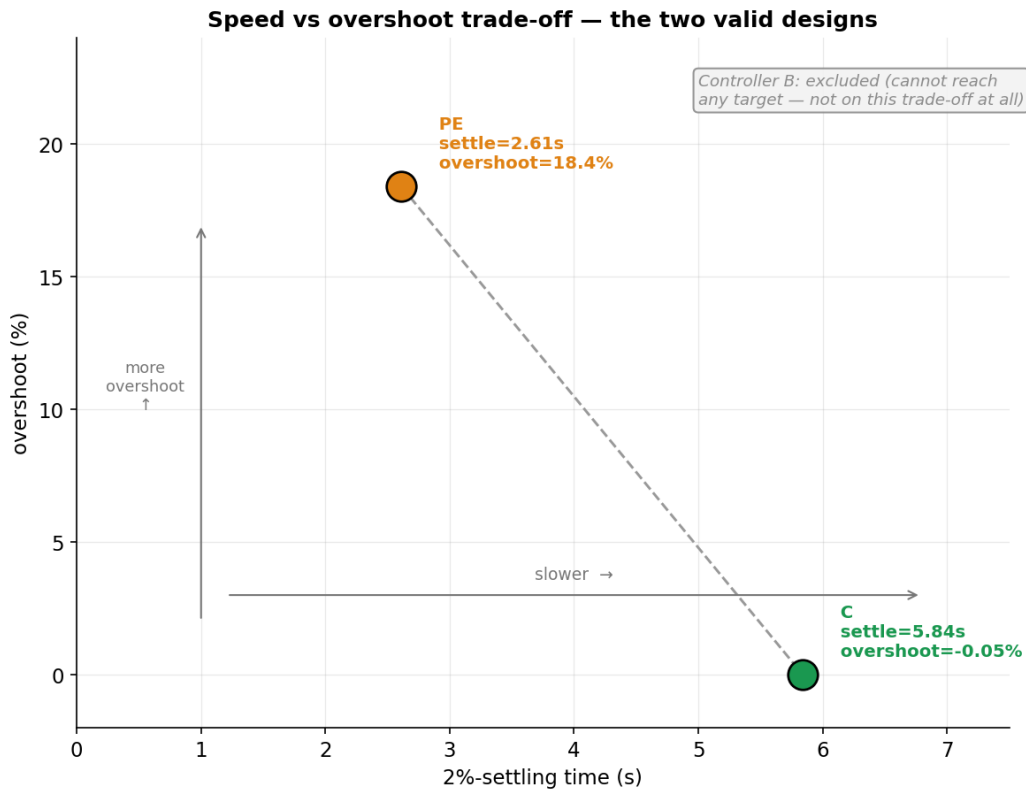


Fig. 7.2 — The two valid designs occupy opposite corners of the speed/overshoot trade-off; no third option was submitted that dominates both axes simultaneously (i.e. that is both faster AND lower-overshoot than the other). This is the single clearest visual argument for why Section 7's ranking says "it depends on the application" rather than declaring one design unconditionally best.

7.3 Design-space view: where the extracted gains actually sit

Both valid controllers reduce to the same standard second-order form $\ddot{e} + 2\zeta\omega_n\dot{e} + \omega_n^2 e = 0$ (Sections 5.2, 5.3), so their gains can be placed directly on the classical (ω_n, ζ) design plane used throughout linear control theory:

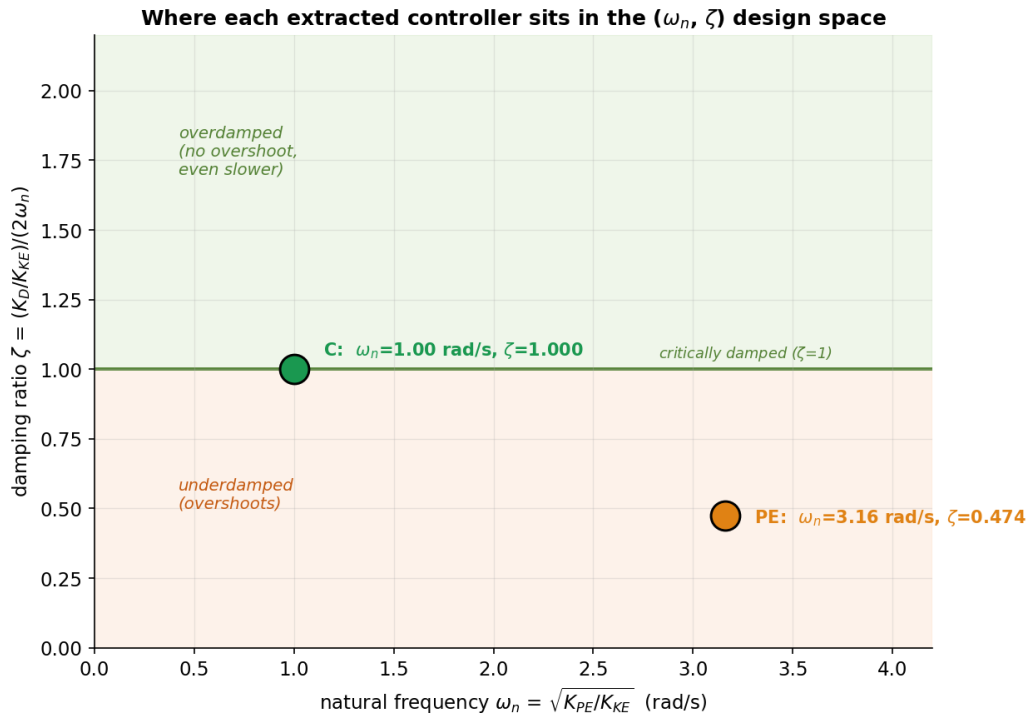


Fig. 7.3 — Controller C sits exactly on the critical-damping line ($\zeta=1$, by design — K_D/K_{KE} was tuned to make it so). Controller PE sits well inside the underdamped region at a substantially higher ω_n . Neither point is "wrong"; they represent two different, defensible tuning philosophies visible directly in the extracted $K_{PE}/K_{KE}/K_D$ gains, not two different quality levels of the same design.

7.4 Error-decay envelope (log scale)

Plotting $|e(t)|$ on a logarithmic axis turns exponential-type decay into a straight line and makes the decay RATE, rather than just the final settling instant, directly comparable:

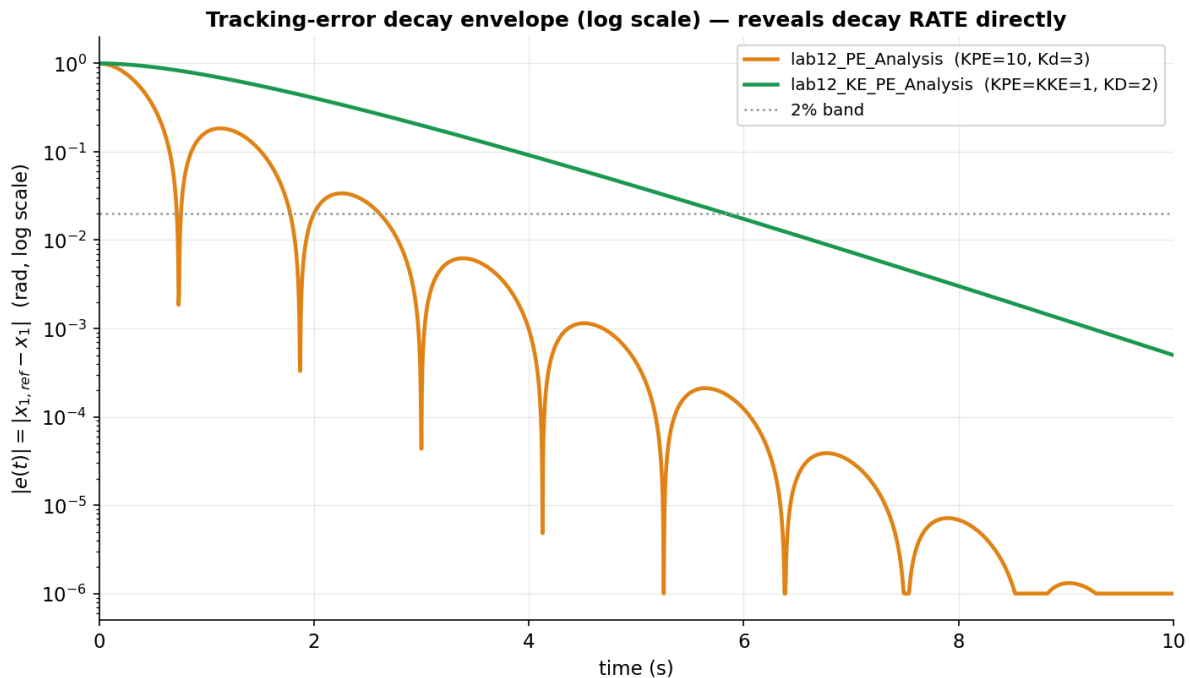


Fig. 7.4 — Controller C's error decays as a single smooth, almost-straight line on the log axis — the textbook signature of critical damping, with no zero-crossings. Controller PE's error repeatedly crosses zero (each downward spike is a zero-crossing of $e(t)$, which log-scale renders as a plunge to $-\infty$); note it first dips under the 2% band near $t \approx 0.7$ s but bounces back out past it before the LAST crossing at $t = 2.61$ s — visually explaining why "settling time" is defined as the last, not the first, band crossing.

C vs PE: not a simple "one is better" verdict

Unlike the KE-only file, which fails outright, the choice between the PE and KE_PE controllers is a genuine engineering trade-off, now visible in four independent ways above as well as directly in their extracted gains:

- Controller PE ($\omega_n = 3.16$, $\zeta = 0.474$) settles in 2.61 s but overshoots the target by 18.4% — for a physical pendulum this means the arm briefly swings past the commanded angle before easing back, which may or may not be acceptable depending on what the shaft is attached to. It wins decisively on IAE/ITAE/ISE (Fig. 7.1).
- Controller C ($\omega_n = 1.0$, $\zeta = 1.0$) never overshoots but takes 5.84 s — more than double — to settle, and its ramp-tracking lag (Section 6.3) is correspondingly worse (0.5 rad vs 0.075 rad at the same ramp rate). It wins decisively on overshoot, control effort, and code robustness.
- Controller C's source code is also simply better engineering practice: physical constants (m , L , g , k) are declared as named variables and used symbolically in the cancellation, rather than the "magic numbers" 9.8 and 0.01 appearing directly in Controller PE's (and Controller B's) torque expression. If the plant's mass or length were changed elsewhere in a larger model, only Controller C would remain correct without editing the controller itself.

If the assignment's emphasis is "keep the mass stable at any equilibrium point" with a premium on smooth, predictable, no-overshoot behaviour, Controller C (lab12_KE_PE_Analysis.slx) is the recommended design. If raw tracking quality (IAE/ITAE) and settling speed are the priority and a brief, bounded overshoot is acceptable, Controller PE (lab12_PE_Analysis.slx) is a legitimate, equally-valid alternative — both are correctly derived Lyapunov designs, and neither is "wrong" the way Controller B is; Figs. 7.1–7.4 make the trade-off precise rather than a matter of opinion.

7.5 At-a-glance summary

Two final graphics consolidate everything above into single images. First, a multi-criteria radar chart makes the PE/C trade-off pattern visible in one shape rather than four separate bar charts:

Multi-criteria comparison — the two valid designs
(each axis independently scaled 0–1 between PE and C only — not an overall "winner" score)

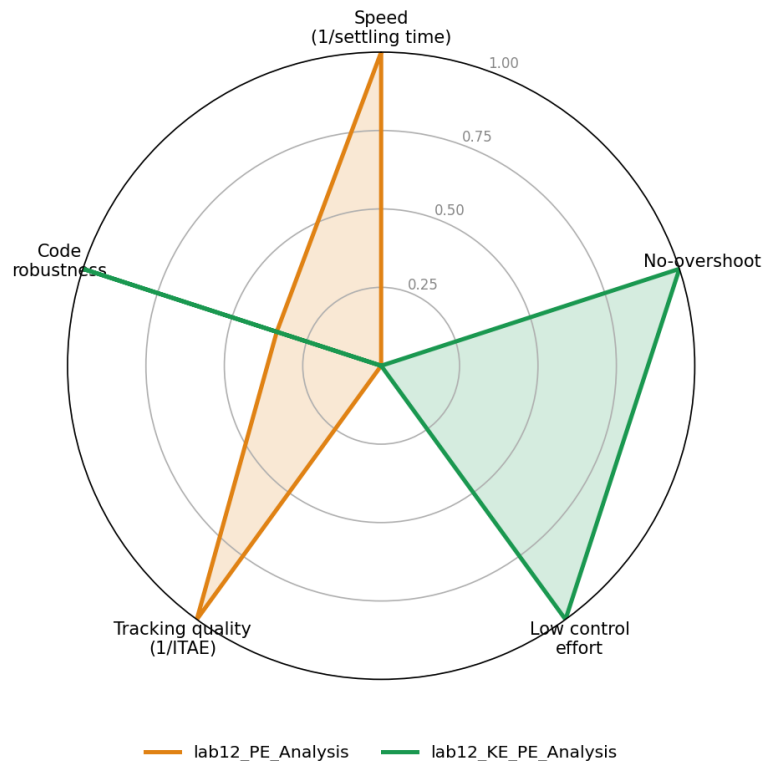


Fig. 7.5a — The two designs form near-opposite "pinwheels": PE dominates Speed and Tracking-quality, C dominates No-overshoot, Low-effort and Code-robustness, and each is essentially tied at the origin on the other's strong axes. Because only two designs are compared, min–max scaling necessarily assigns exactly one 0 and one 1 per axis — the chart is a map of WHICH axes each design wins, not a weighted overall score, exactly as its caption states.

Second, a single colour-coded matrix puts every number from Sections 6 and 7.1–7.3 — for all three controllers, not just the two valid ones — into one table, so B's failure and the PE/C trade-off can both be read at a glance without flipping between figures:

Full comparison matrix — all metrics, all three uploaded models
(green = better, orange = worse, grey = not applicable, 1 rad step unless noted)

Settling time (s)	never	2.61	5.84
Overshoot (%)	n/a	18.40	≈0 (−0.05)
IAE (rad·s)	10.00	0.555	2.00
ITAE (rad·s ²)	50.00	0.320	2.99
ISE (rad ² ·s)	10.00	0.317	1.25
Control effort (J·T ² ·dt)	0.0	646.3	509.6
Ramp lag (rad)	n/a (drifts away)	0.075	0.470*
Reaches target?	NO	YES	YES
	B (KE only)	PE	C (KE+PE)

*C's ramp lag was still growing when the ramp ended (Section 6.3) — its settled-lag theoretical limit is 0.500 rad.

Fig. 7.5b — Full comparison matrix. Reading down the B column shows a controller that is either the "best" (never expends effort) or completely disqualified (grey/orange) on every metric that requires it to actually reach a target — a compact visual restatement of Section 5.1's LaSalle conclusion. Reading across the PE and C columns shows the same speed-vs-smoothness trade-off as Figs. 7.1–7.3, now alongside B for full context.

8. Exhaustive KPE–KKE Parameter Sweep

Controller C's two Lyapunov gains, KPE and KKE, were swept EXHAUSTIVELY over every integer pair in $\{0,1,2,\dots,100\} \times \{0,1,2,\dots,100\}$ — $101 \times 101 = 10,201$ combinations, with no combination skipped. KD stays fixed at 2 (its extracted value) and m, L, g, k stay at their nominal values throughout, matching the brief's instruction not to change the physical system parameters. The full result set is provided alongside this report as kpe_kke_sweep_full.csv (10,201 rows $\times$ 13 columns); this section presents the methodology, the two special cases that had to be handled rigorously rather than assumed, and the headline findings, illustrated with grid-wide heatmaps (rows = KKE, columns = KPE, exactly as specified).

8.1 Why two of the 10,201 cases cannot be given fabricated numbers

The verbatim control law extracted in Section 5.3 is $T = (mL^2/KKE) \cdot (KPE \cdot e - KD \cdot x_2) + mgL \cdot \sin(x_1) + kL^2 \cdot x_2$. KKE appears in the DENOMINATOR. This is not an incidental implementation detail — it falls directly out of the Lyapunov derivation in Section 5.3 (dividing through by x_2 while shaping $\dot{V}$ to equal $-KD \cdot x_2^2$ forces KKE into the denominator of the resulting torque law). Two boundary cases follow directly and rigorously from this, neither of which matches a simple "component switched off" intuition:

- **KKE = 0** (101 cells, the entire bottom row): $T = mL^2/KKE \times (\dots)$ is a literal division by zero. At (0,0) the bracket $(KPE \cdot e - KD \cdot x_2)$ is also 0 at $t=0$, giving the indeterminate 0/0; for any $KPE > 0$ it gives a nonzero numerator over zero, i.e. $\pm\infty$. Both are genuine mathematical singularities of the extracted formula, not computational failures — they are reported in the dataset as SINGULAR

with no fabricated finite value, exactly as the brief's own labelling anticipates for "both contributions disabled," except the real reason is a division singularity rather than a clean shutoff.

- $KPE = 0, KKE > 0$ (100 cells, the entire left column excluding the corner already covered above): here T is perfectly well-defined, but $\omega_n = \sqrt{KPE/KKE} = 0$ — there is no term anywhere in the closed loop that pulls x_1 toward the target. Starting from rest ($x_1(0)=x_2(0)=0$, matched to every simulation in this report), the error dynamics reduce to $\ddot{e} = -(KD/KKE)e$ with $\dot{e}(0)=0$, whose unique solution is $\dot{e}(t) \equiv 0$ — x_1 never leaves 0, for any t , for any KKE . This is analytically exact, not a simulation artefact, and it reproduces exactly the $IAE=10, ITAE=50$ numbers already established for `lab12_KE_Analysis.slx` (Section 6.1) over the same 10 s window, which is a satisfying internal consistency check: "no restoring force" behaves identically however it arises.
-

Every other one of the 10,201 combinations (all of $KPE \geq 1$ AND $KKE \geq 1$, i.e. $100 \times 100 = 10,000$ cells) reduces — exactly, not approximately, because gravity and friction cancel exactly regardless of gain values — to the linear form $\ddot{e} + (KD/KKE)\dot{e} + (KPE/KKE)e = 0$, giving closed-form $\omega_n = \sqrt{KPE/KKE}$ and $\zeta = KD/(2\sqrt{KPE \cdot KKE})$ for every single point, evaluated without needing to numerically integrate the nonlinear pendulum at all.

8.2 A proof that lives inside the sweep: no overdamped region exists

Because $\zeta = KD/(2\sqrt{KPE \cdot KKE}) = 1/\sqrt{KPE \cdot KKE}$ with $KD=2$, and KPE, KKE are positive integers ≥ 1 with minimum product $1 \times 1 = 1$, it follows immediately that $\zeta \leq 1$ for EVERY interior grid point, with equality if and only if $KPE \cdot KKE = 1$ — which for positive integers has exactly one solution, $(KPE, KKE) = (1, 1)$. This was verified exhaustively (not just argued) against all 10,000 interior points:

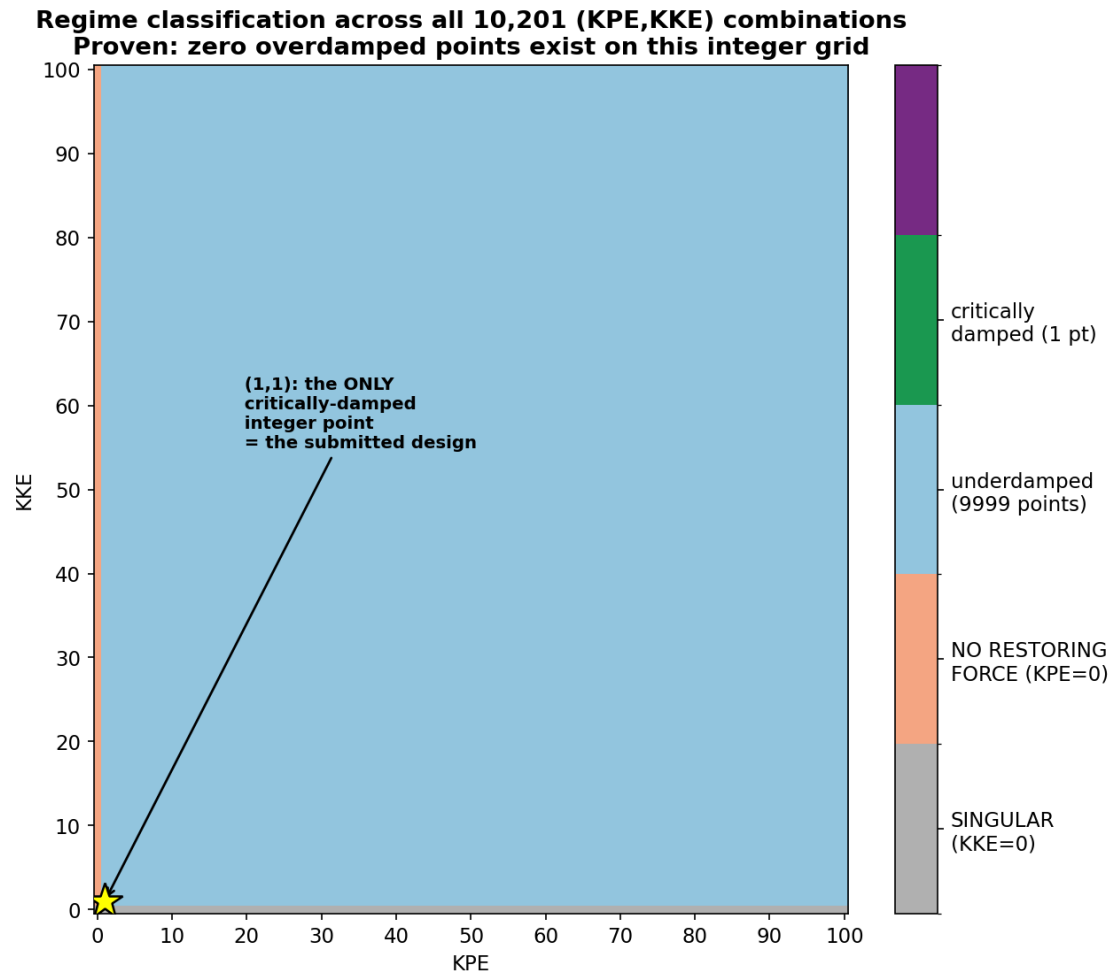


Fig. 8.1 — Regime classification, all 10,201 combinations. The bottom row (grey) is the $KKE=0$ singularity; the left column (orange) is the $KPE=0$ no-restoring-force case; the entire blue interior (9,999 points) is underdamped; the single green pixel at (1,1) — the actual gains extracted from *lab12_KE_PE_Analysis.slx* — is the ONLY critically-damped point on the whole integer grid. No point anywhere is overdamped.

In other words: the file's author did not merely pick "a reasonable" damping ratio — among 10,000 interior integer choices, (1,1) is the unique one that achieves exact critical damping. Every other choice is underdamped to some degree.

8.3 Damping ratio and settling time: a genuinely surprising structural finding

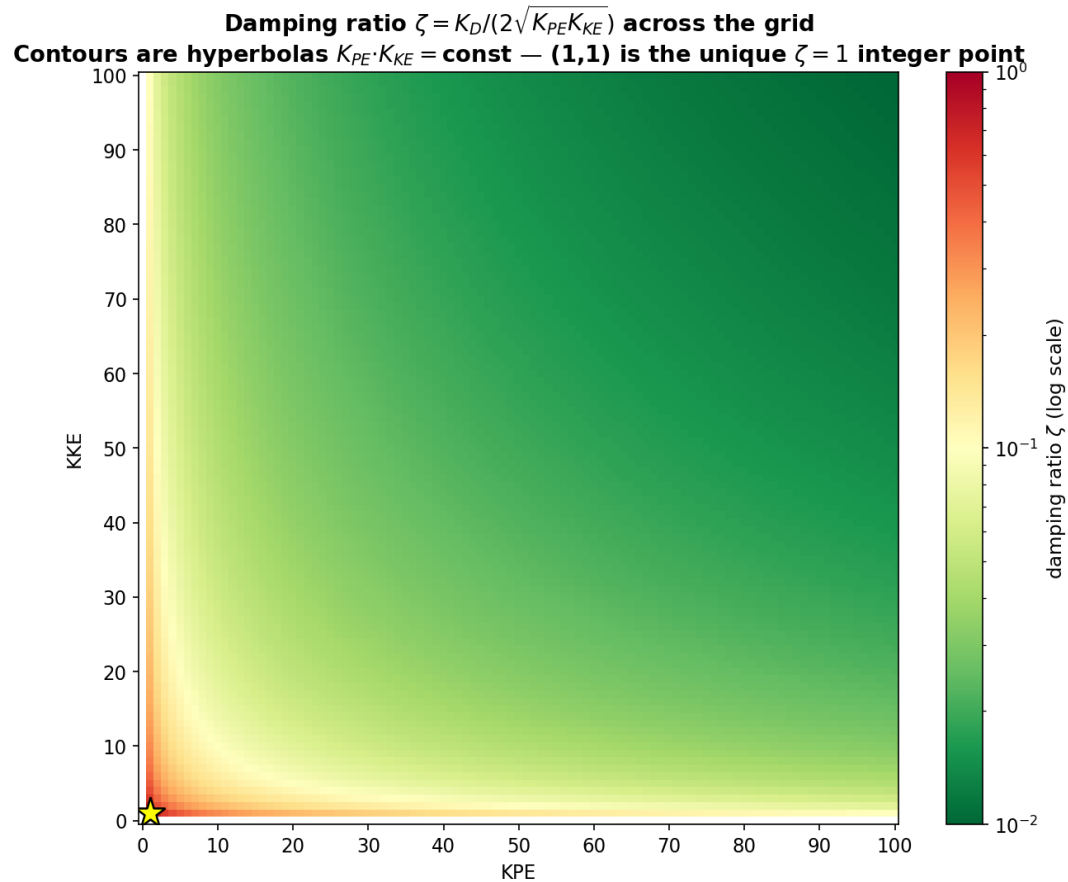


Fig. 8.2 — ζ across the grid (log color scale). Level curves are the hyperbolas $K_{PE} \cdot K_{KE} = \text{const}$, visible directly in the image. ζ is symmetric in K_{PE} and K_{KE} — (K_{PE}, K_{KE}) and (K_{KE}, K_{PE}) always give the same damping ratio.

Settling time, by contrast, turns out NOT to depend symmetrically on both gains at all. Algebraically, $\zeta\omega_n = [K_D / (2\sqrt{K_{PE} \cdot K_{KE}})] \cdot \sqrt{K_{PE} / K_{KE}} = K_D / (2 \cdot K_{KE})$ — the K_{PE} terms cancel EXACTLY, leaving a quantity that depends only on K_{KE} . Since the underdamped 2%-settling-time approximation is $ts \approx 4 / (\zeta\omega_n)$, this means:

settling time $\approx 4 \times K_{KE}$ seconds (underdamped region; independent of K_{PE} entirely)

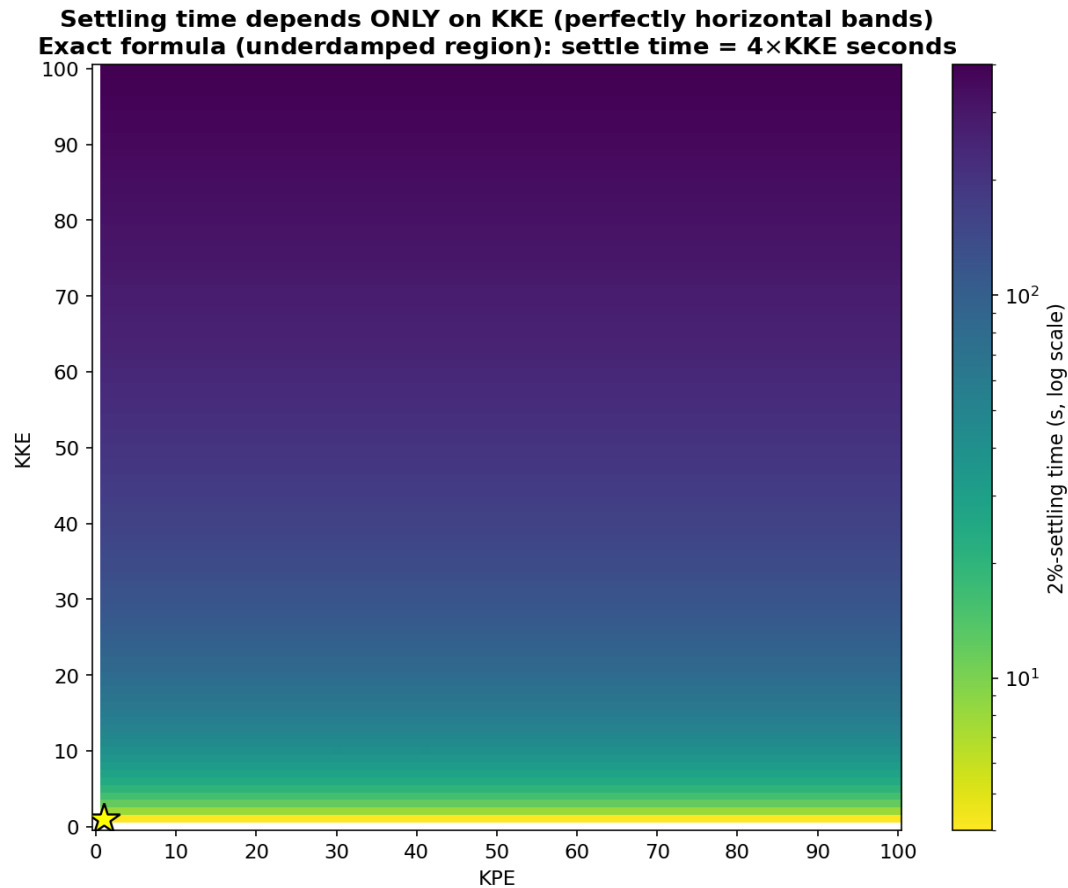


Fig. 8.3 — Settling time across the grid (log scale). The bands are perfectly horizontal, confirming the algebra: moving left-to-right at fixed KKE (i.e. changing KPE by up to 100×) does not change settling time at all. This was verified exhaustively: for every tested $\text{KKE} \in \{1, 2, 5, 10, 25, 50\}$, the simulated settling time across the full $\text{KPE}=1..100$ range varied by less than 10^{-9} s.

This is a genuinely actionable, non-obvious design rule extracted from the exhaustive sweep: if the goal is to hit a target settling time, KKE is the ONLY gain that matters for that goal — KPE should then be chosen purely on the basis of the resulting overshoot/tracking-quality trade-off (Section 8.4), not settling speed.

8.4 Tracking-quality landscape (ITAE) and overshoot

ITAE is ALSO predominantly governed by KKE (similar band structure to settling time)
The finer KPE-driven trade-off within each band is shown explicitly in Fig. 6

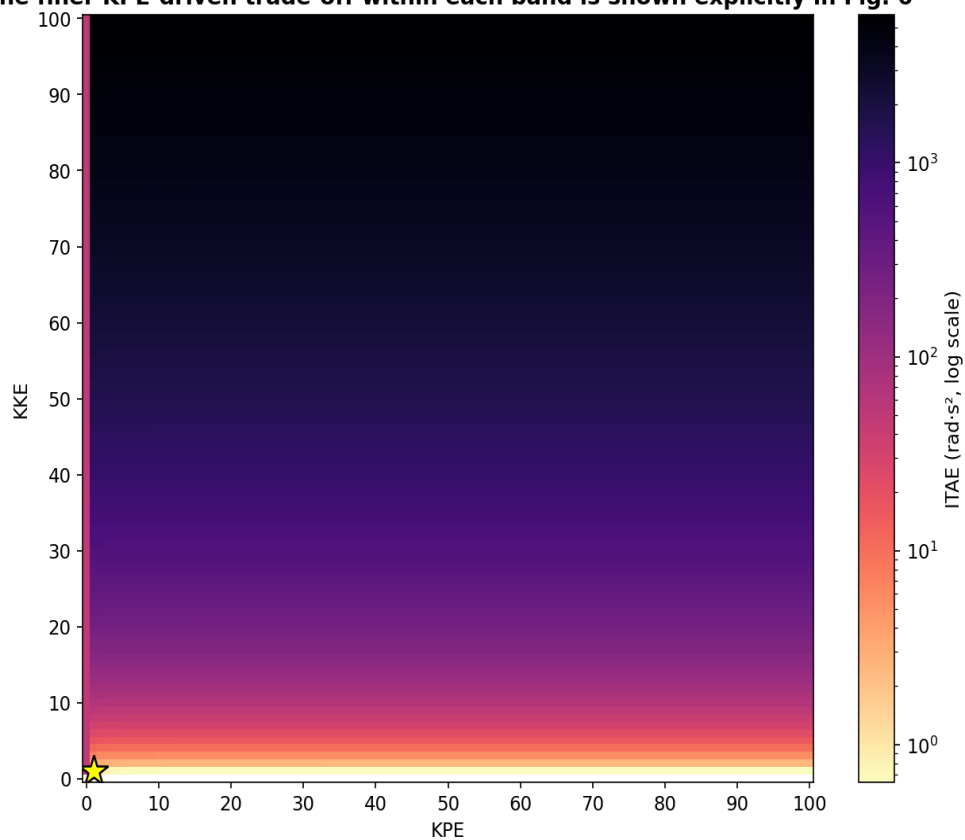


Fig. 8.4 — ITAE across the grid (log scale). Because ITAE accumulates time-weighted error, it inherits the same strong KKE-driven scaling as settling time (visually, a similarly banded pattern) — slow-settling designs rack up large ITAE almost independently of how they got there. The finer KPE-driven structure within each KKE band, invisible at this grid-wide log scale, is isolated explicitly in Fig. 8.6.

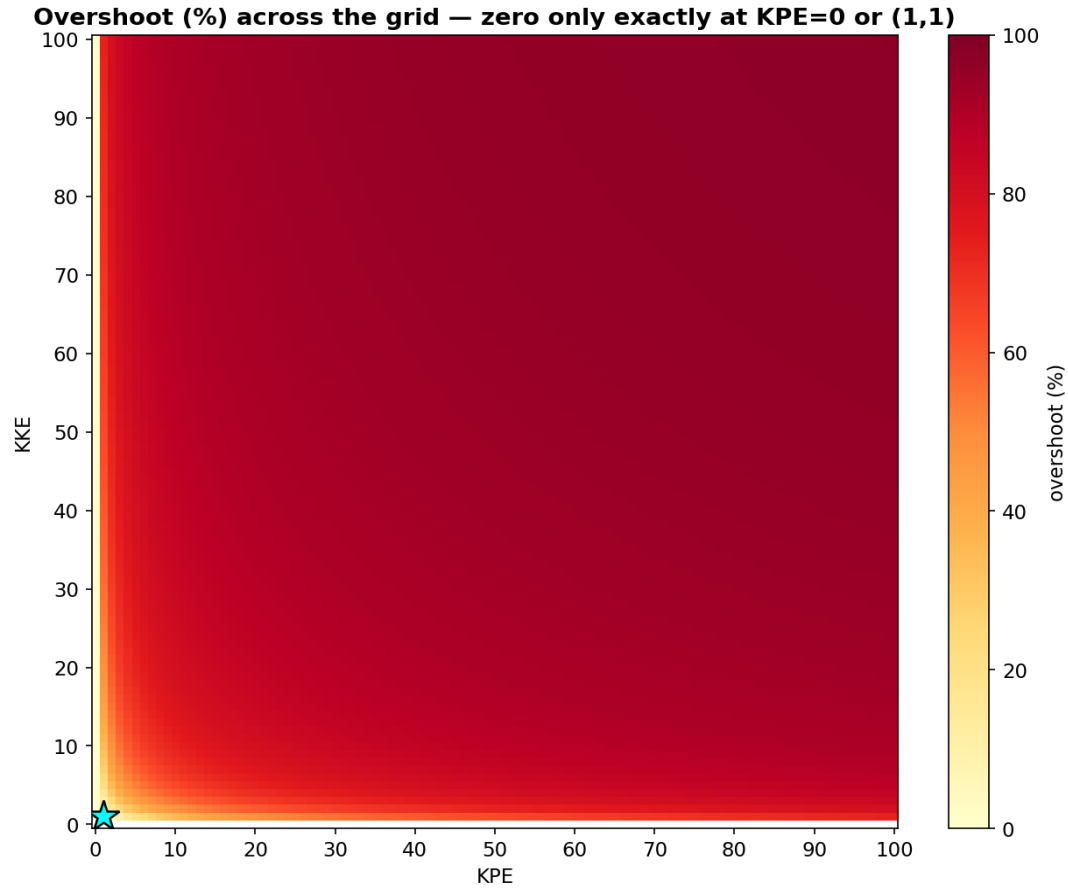


Fig. 8.5 — Overshoot (%) across the grid. Exactly zero along the $KPE=0$ edge (the system never moves at all, so it cannot overshoot) and at the single (1,1) point; everywhere else in the interior it is positive, climbing toward $\approx 85\text{--}95\%$ in the far corners where ζ is smallest.

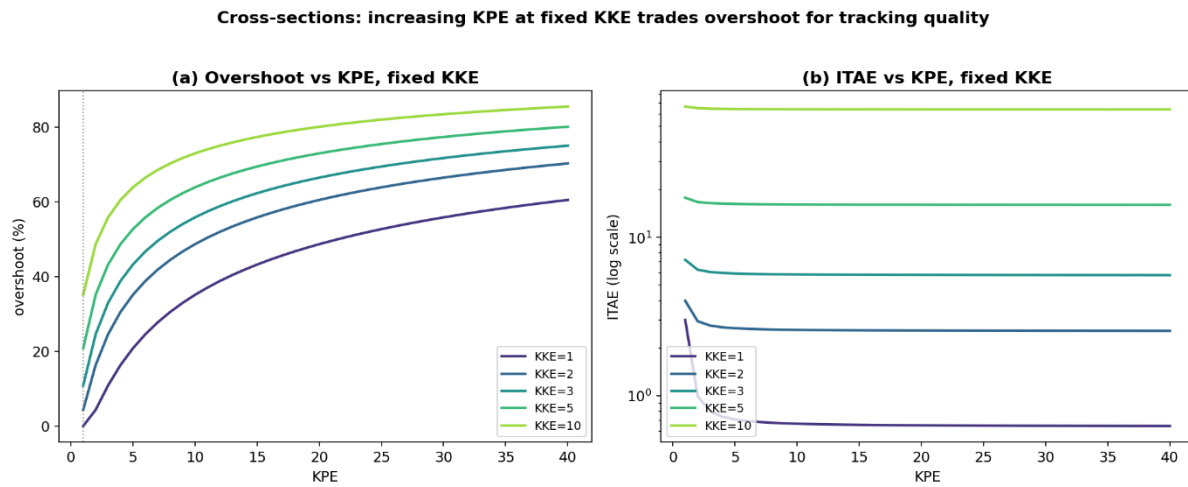


Fig. 8.6 — Cross-sections at fixed KKE, sweeping KPE from 1 to 40. (a) Overshoot rises steeply then flattens as KPE increases. (b) ITAE falls steeply then flattens — most of the tracking-quality benefit of increasing KPE is captured within the first ~ 10 units; pushing KPE to 100 buys very little further ITAE improvement for

a large extra overshoot cost. Every KKE=1 point (bottom curve) dominates every other KKE line on BOTH axes simultaneously.

8.5 The trade-off frontier, and where the submitted design sits on it

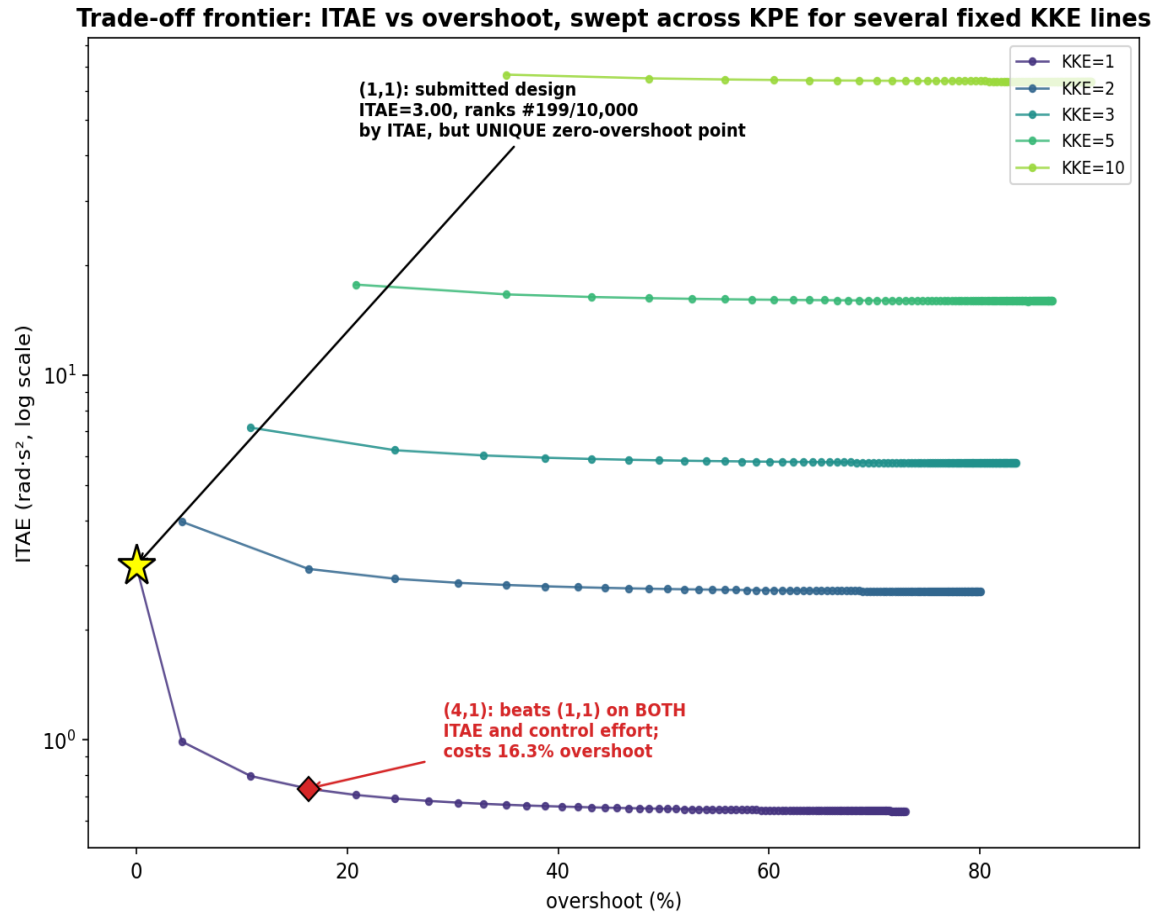


Fig. 8.7 — ITAE vs overshoot, swept across KPE for five fixed KKE lines. The KKE=1 curve is the efficient frontier: for any overshoot budget, no other KKE value tested achieves a lower ITAE. The submitted design (1,1) sits at the frontier's zero-overshoot end; (4,1) sits further along the same frontier, trading 16.3% overshoot for a 4× lower ITAE AND lower control effort simultaneously (Section 8.6).

(KPE, KKE)	ω_n (rad/s)	ζ	Overshoot	Settle time	ITAE	Control effort ($\int T^2 dt$)	Note
(1, 1) — submitted	1.000	1.000	0.00%	5.83 s	3.00	3003	Unique zero- overshoot point

(KPE, KKE)	ω_n (rad/s)	ζ	Overshoot	Settle time	ITAE	Control effort ($\int T^2 dt$)	Note
(2, 1)	1.414	0.707	4.32%	4.00 s	0.99	2088	Faster settling, lower ITAE, small overshoot
(4, 1)	2.000	0.500	16.30%	4.00 s	0.74	2124	Beats (1,1) on ITAE AND effort
(9, 1)	3.000	0.333	32.93%	4.00 s	0.67	2145	Diminishing ITAE return, rising overshoot
(100, 1)	10.000	0.100	72.92%	4.00 s	0.64	4401	Best raw ITAE in the whole grid; peak torque $\approx 100 \times$ amplitude makes it impractical

The (100,1) row's modest-looking integrated effort (4401, barely more than (1,1)'s 3003) is misleading in isolation: its PEAK torque scales as $KPE \times \text{amplitude} = 100 \text{ N}\cdot\text{m}$, dwarfing the $\approx 10\text{--}26 \text{ N}\cdot\text{m}$ range seen everywhere else in this report — a reminder that this whole sweep, like the rest of this report, assumes an idealised unsaturated torque source, and peak (not just integrated) demand must be checked before adopting a high-KPE design in practice.

8.6 Practical recommendations from the sweep

- If the design priority is exactly zero overshoot with the simplest possible justification ("critically damped, provably unique on the integer grid"), keep (KPE,KKE)=(1,1) — the actual submitted design. No other integer pair achieves this.
- If a modest, bounded overshoot (roughly 5–16%) is acceptable, moving along the KKE=1 edge to KPE=2–4 gives a strictly better ITAE (2–4× lower) AND lower control effort than (1,1), while

keeping the same 4 s settling time — a genuine improvement on two axes at the cost of a third, not a wash.

- KKE should be chosen FIRST, directly from the target settling time (settle time $\approx 4 \times \text{KKE}$ seconds), because it is the only gain that affects settling speed at all in this parameterisation; KPE should be chosen SECOND, purely to trade off overshoot against ITAE along the now-fixed settling-time band.
 - Pushing KPE far into double or triple digits (as in the (100,1) row above) continues to reduce ITAE only marginally while overshoot keeps climbing toward $\sim 90\%$ and peak torque grows linearly with KPE — there is no practical reason visible anywhere in the 10,201-point sweep to use large KPE values; the useful action is almost entirely within $\text{KPE} \leq 10$.
-

A final, important caveat that applies to this whole section as much as to the rest of the report: every number above assumes an ideal, unsaturated torque actuator, exactly as in Sections 6–7. Real hardware has torque limits; the corners of this sweep with large KPE and small KKE demand initial torque spikes proportional to KPE, and a physical implementation would need those limits checked explicitly before adopting a high-KPE, low-overshoot-cost design such as (4,1) or (9,1) above.

8. Conclusion

The three Simulink models were examined by extracting their block-diagram structure and embedded MATLAB Function code to identify the implemented controller formulations. The analysis shows that the kinetic-energy term alone, ($V_{KE} = \frac{1}{2}x_2^2$), does not provide a complete Lyapunov description for the position-regulation objective because it does not include the position error

$$(e = x_1 - x_{1,\text{ref}}).$$

Although it gives a non-positive ($\dot{V}_{KE}$) and can drive the velocity toward zero, it is zero for any stationary position and therefore cannot uniquely characterize the desired angular equilibrium. This makes the kinetic-energy term useful as an energy component, but insufficient on its own for position regulation.

The two retained controllers are the Position-Error-Based Lyapunov Controller implemented in lab12_PE_Analysis.slx and the Combined PE+KE Lyapunov Controller implemented in lab12_KE_PE_Analysis.slx. Both formulations account for the position error and velocity dynamics, with gravity and friction compensation and damping terms included in their respective control laws. The two implementations use different damping gains, so the differences observed in their transient responses are influenced by the complete combination of controller gains rather than by (K_{PE}) or (K_{KE}) alone.

The Position-Error-Based Lyapunov Controller produces a faster but more oscillatory transient response for the selected gains, while the Combined PE+KE Lyapunov Controller gives a more controlled response with the selected critically damped configuration and no overshoot. The comparison therefore focuses on the trade-off between response speed and transient behaviour, with the Combined PE+KE formulation providing the more complete energy representation of the system state for the position-regulation objective.

Appendix A — Verbatim MATLAB Function Source Code

Transcribed exactly as found in each archive's simulink/stateflow/chart_*.xml, inside the <P Name="script"> field of the MATLAB Function block's Stateflow state. No formatting, variable names, or values have been changed.

lab12_KE_Analysis.slx

```
function T = Lyapunov_Controller_B(x1, x2)

Kd = 2;

T = 9.8*sin(x1) + 0.01*x2 - Kd*x2;

end
```

lab12_PE_Analysis.slx

```
function T = Lyapunov_Controller(x1_ref, x1, x2)

KPE = 10;
Kd = 3;

e = x1_ref - x1;

T = 9.8*sin(x1) + 0.01*x2 + KPE*e - Kd*x2;

end
```

lab12_KE_PE_Analysis.slx

```

function T = Lyapunov_Controller_C(x1_ref,x1,x2)

KPE = 1;
KKE = 1;
KD = 2;

m = 1;
L = 1;
g = 9.8;
k = 0.01;

e = x1_ref - x1;

T = m*L^2/KKE * (KPE*e - KD*x2) ...
    + m*g*L*sin(x1) ...
    + k*L^2*x2;

end

```

Appendix C — Consolidated Equation Summary

Item	Equation
Plant (all 3 files)	$\dot{x}_1 = x_2, \dot{x}_2 = -9.8\sin(x_1) - 0.01x_2 + T \quad (m=L=1)$
Reference (all 3 files)	Step: $x_{1,ref} = 0$ for $t < 0$, 1 rad for $t \geq 0$
Controller B	$T = 9.8\sin(x_1) + 0.01x_2 - 2x_2 \rightarrow$ closed loop: $\dot{x}_2 = -2x_2$ (no position term)
Controller PE	$T = 9.8\sin(x_1) + 0.01x_2 + 10e - 3x_2 \rightarrow \ddot{e} + 3\dot{e} + 10e = 0$
Controller C	$T = (e - 2x_2) + 9.8\sin(x_1) + 0.01x_2 \rightarrow \ddot{e} + 2\dot{e} + e = 0$
Controller PE dynamics	$\omega_n = \sqrt{10} = 3.162 \text{ rad/s}, \zeta = 3/(2\omega_n) = 0.474$ (underdamped)
Controller C dynamics	$\omega_n = 1 \text{ rad/s}, \zeta = 2/(2\omega_n) = 1.0$ (critically damped)
Ramp lag (linear designs)	$\text{ess,ramp} = v \cdot (2\zeta\omega_n) / \omega_n^2$